

CT216: Introduction to Communication Systems

Course Placement a core course for the second year BTech ICT and ICT+CS students

Course Format three Lecture hours and two Lab hours per week

Course Content The course CT216 introduces the students to the various engineering techniques that empower the modern communication systems. The course demonstrates application of elegant mathematical principles to practical engineering design of the communication systems. The students gain hands-on skills on these topics by performing software laboratory experiments throughout the semester and by completing one project in the semester. The specific topics covered are information theoretic limits and power versus bandwidth trade-off in a communication system, introduction to signals and systems, digital and analog modulation schemes, design of receivers and demodulators, channel coding schemes, seven layer OSI model, etc.

Textbook U. Madhow, Introduction to Communication Systems, Cambridge University Press, 2018

Reference Books The following books are used as the reference:

1. L. Couch, Digital and Analog Communication Systems, Pearson, 2012
2. D. J C MacKay, Information Theory, Inference, and Learning Algorithms, Cambridge University Press, 2003
3. John G. Proakis and Masoud Salehi, Digital Communications, Fifth Edition, McGraw-Hill, 2014
4. Stephen Wicker, Error Control Systems for Digital Communication and Storage, Prentice-Hall, 1994

Grading Policy The assessment method will be as follows:¹

- Two midterm exams: 22.5% each
- Final exam: 26%
- Labs:² 15%
- Project:³ 10%
- Quizzes: 4%
- Attendance: will follow Institute's attendance policy

Course Outcomes this course introduces the students to the engineering knowledge (P1) of the mathematics and engineering fundamentals of the communication systems. The students engage in the analysis (P2) and design (P3) of the basic modules of a communication system. As part of the Lab and the Project, the students conduct investigations

¹This is tentative and subject to change.

²Students will be tested throughout the semester if the project, the lab and the homework that they submit is their own work, which they fully understand.

³Students will be evaluated on their proactiveness and active participation in their project group, and understanding of the project submission.

of the complex design and analysis problems (P4) and make use the modern tools (P5) such as Matlab and Python. Through the Lab and the Project works that is team-based and require a formal presentation by the students, they acquire the team-work (P9) and the technical communication (P10) skills.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
✓	✓	✓	✓	✓				✓	✓		